CMSC 603 High-Performance Distributed Systems

CUDA image processing



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Examples of using CUDA + OpenCV for parallel image processing

1. Convert an image from BGR to grayscale



See: **BGRtoGrayScale.cu**

Pixels on a color image are BGR represented as

```
struct uchar3 {
    unsigned char x, y, z;
};
```

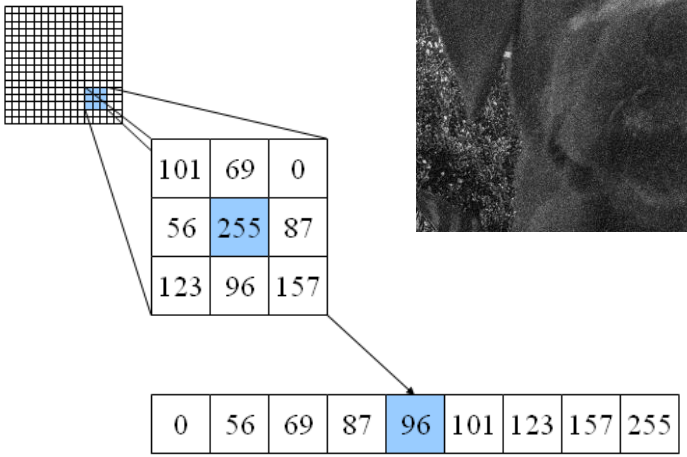
Pixels on a grayscale image are represented as

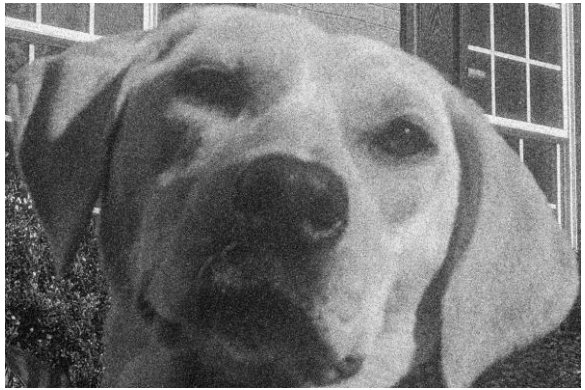
unsigned char [0 – 255]

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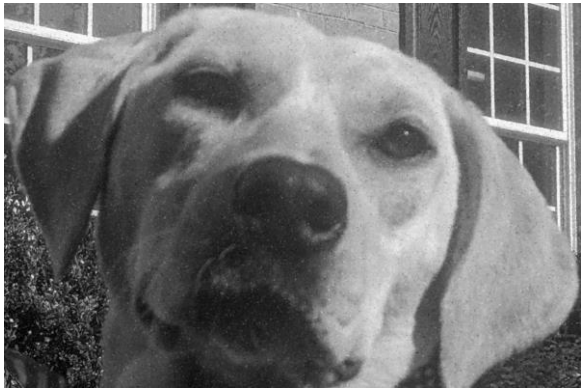
2. Median filter to remove noise from image

See: **MedianFilter.cu**









Important not to overflow image size!

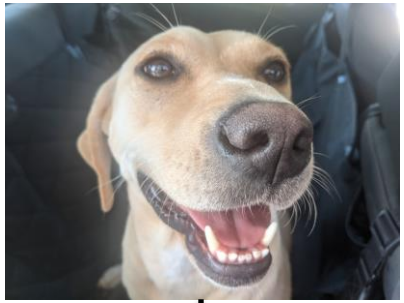
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3. Hide secret image inside an image using LSB-steganography See: **LSBsteganography.cu**

Original



Secret

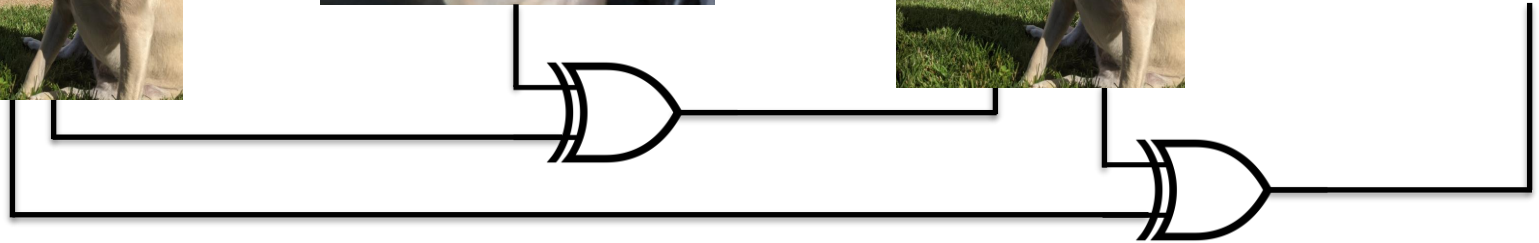


Original + Secret



Extracted secret





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